Power BI Assignment 2 – DAX & Data Visualization

E-Commerce Sales Analysis

1. Transformed and loaded the given data using Power query

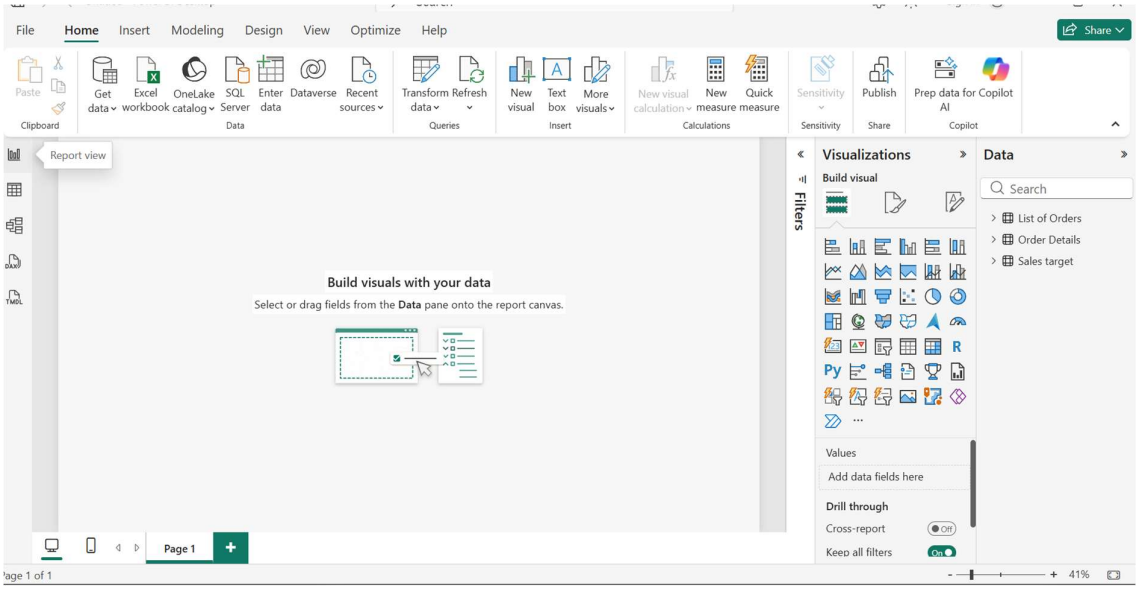
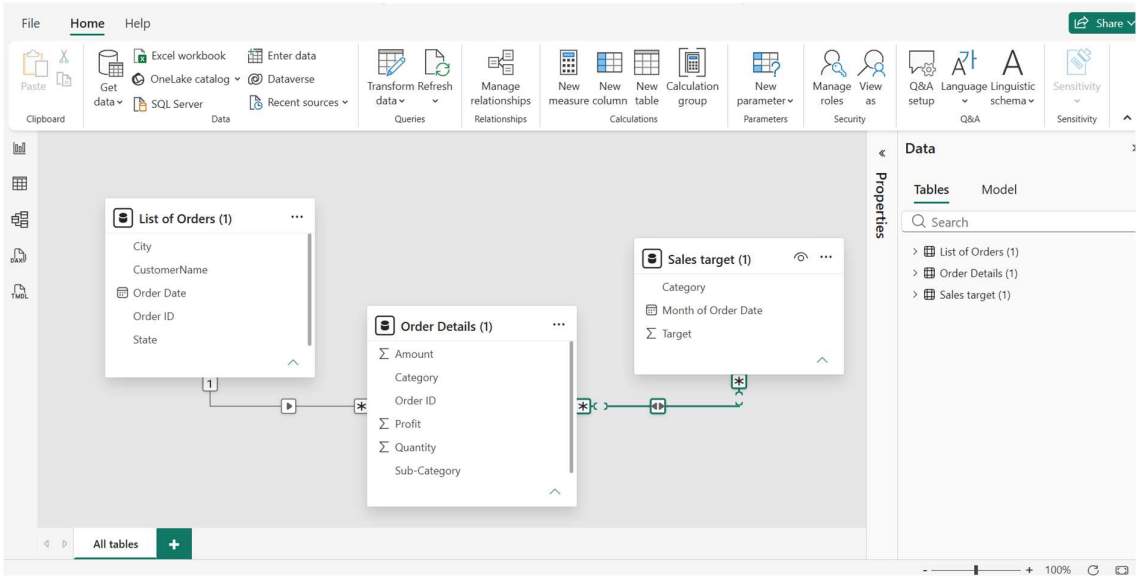
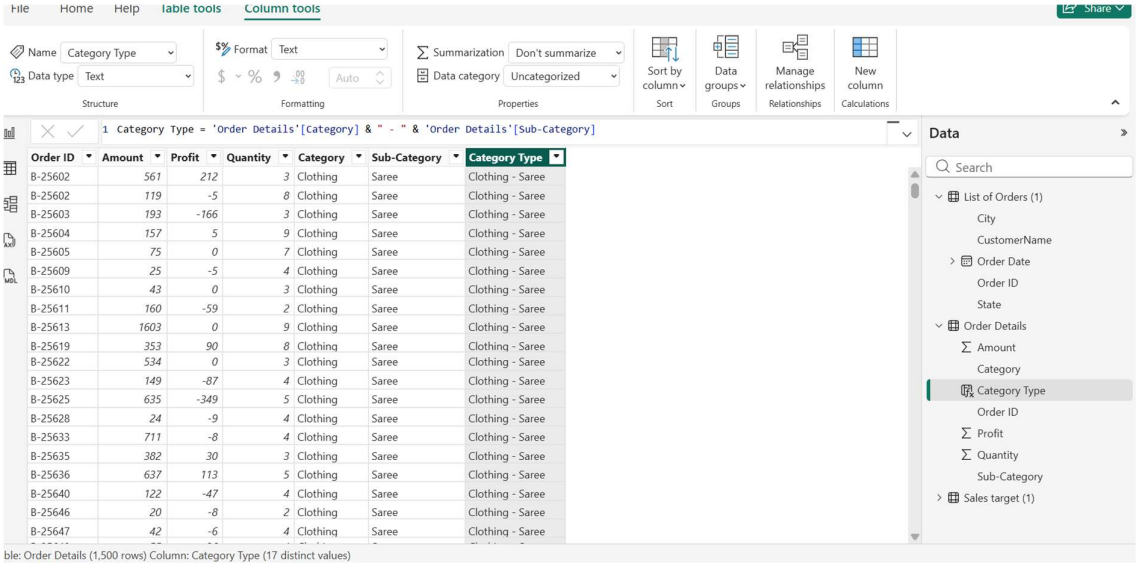


Table Relationship has been established.

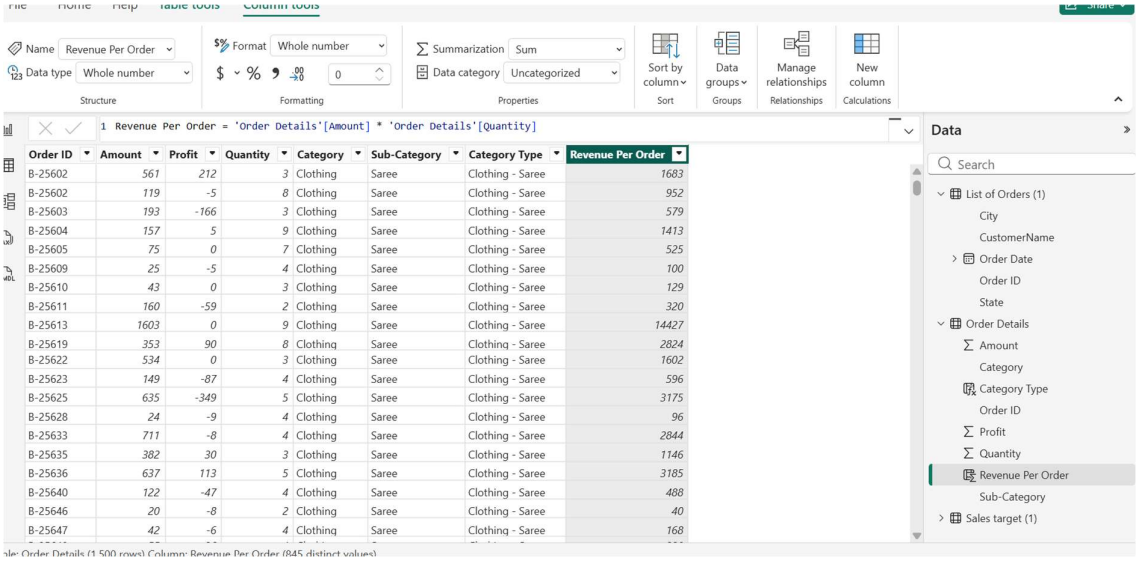


Calculated Columns:

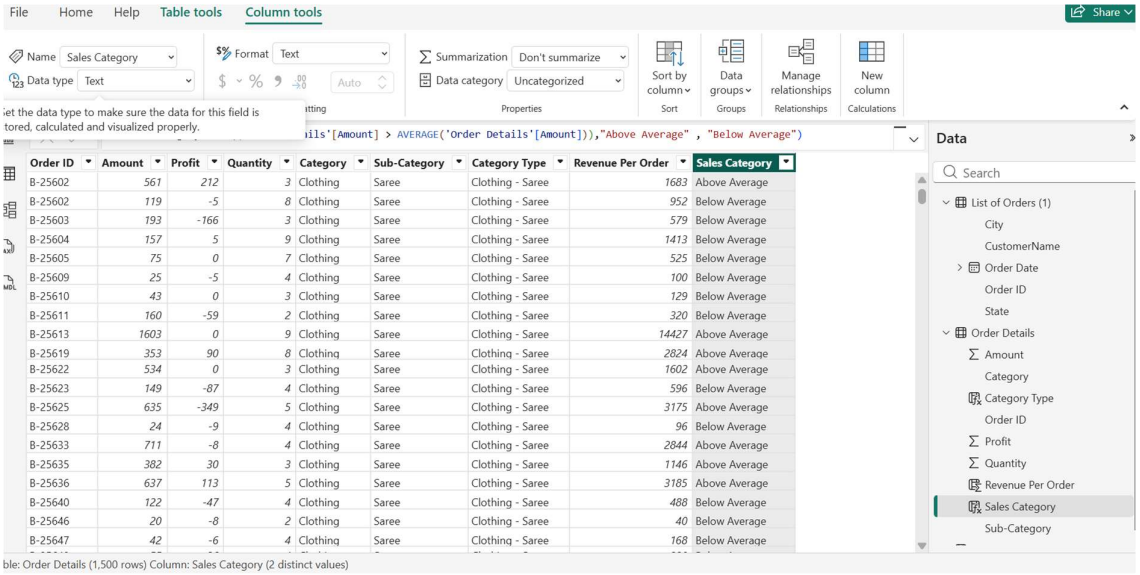
Added a new column named **“Category Column”** by combining Category and Sub-category column



Calculated **Revenue Per Order(considering this as sales)** in Order Details table as below

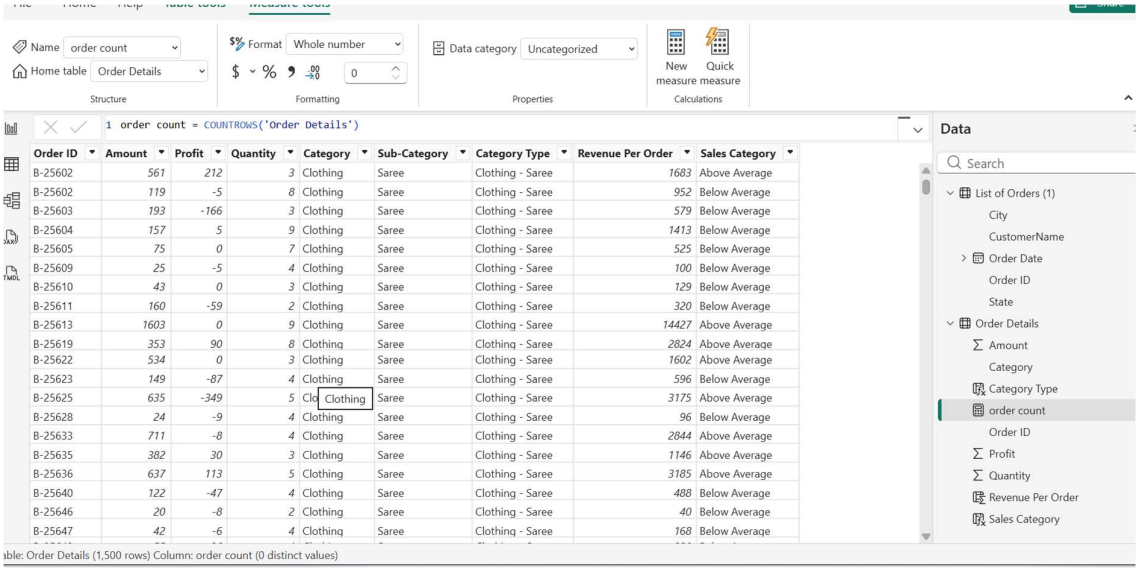


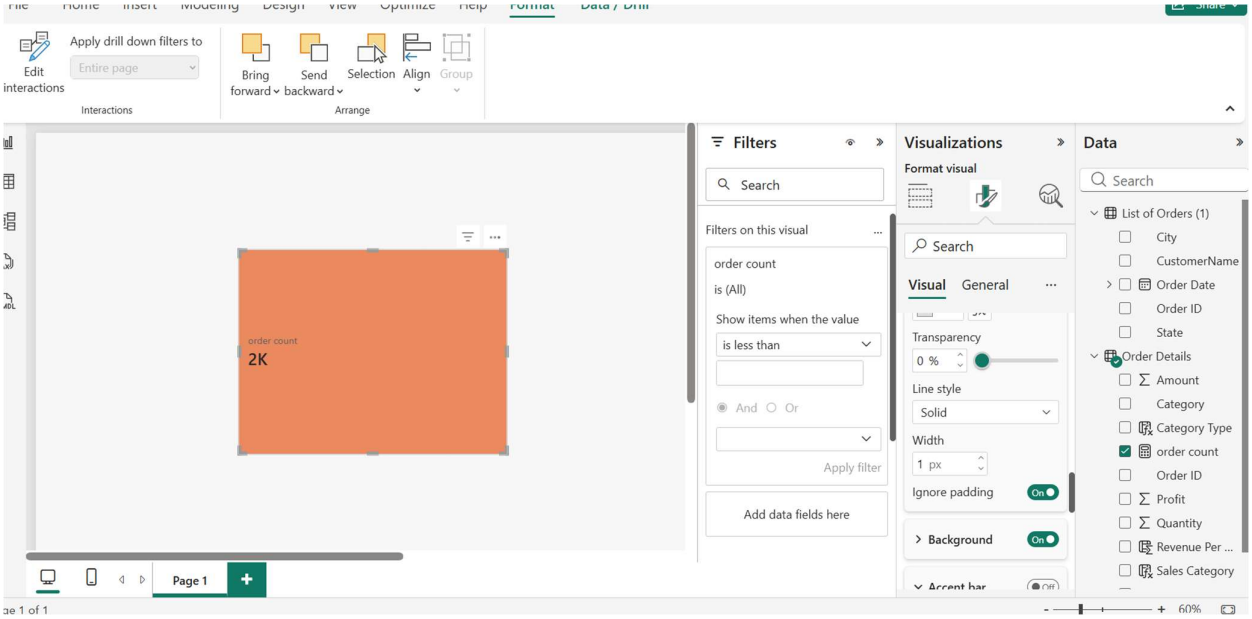
Added a calculated column named **‘Sales Category’** in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.



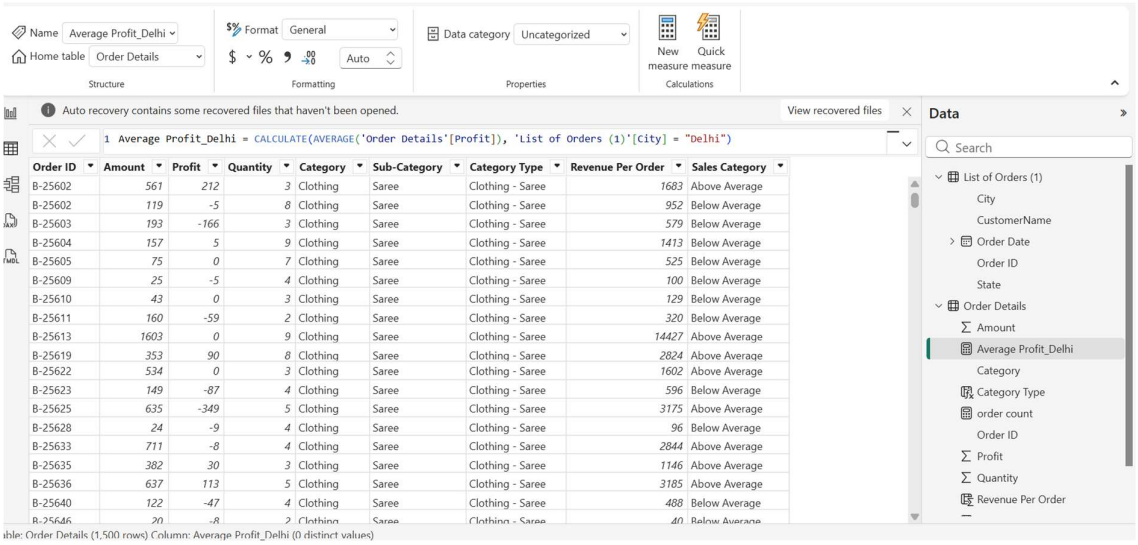
Calculated Measures:

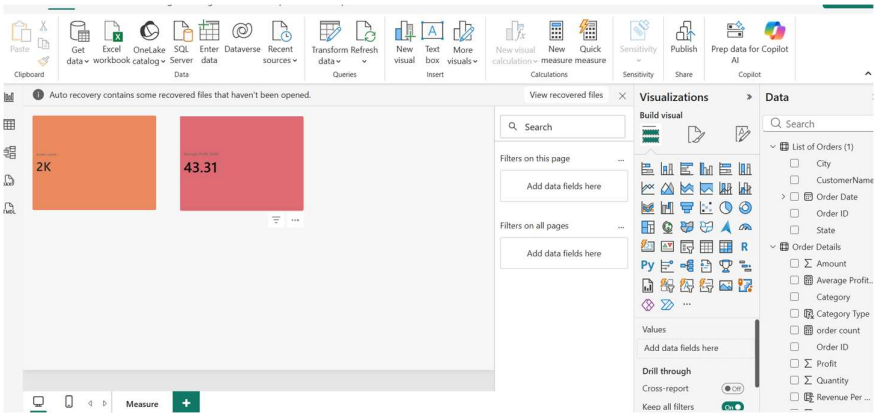
Calculate Order Count: Calculated a measure to count the total number of orders in the Order Details table and displayed the result in a card.



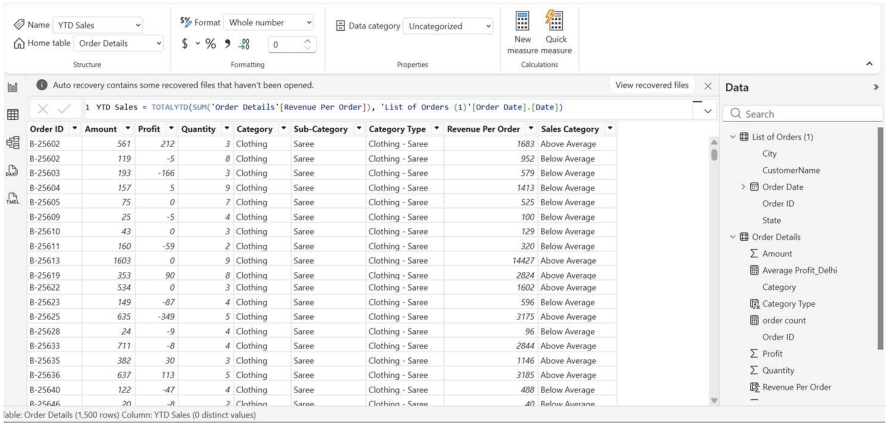


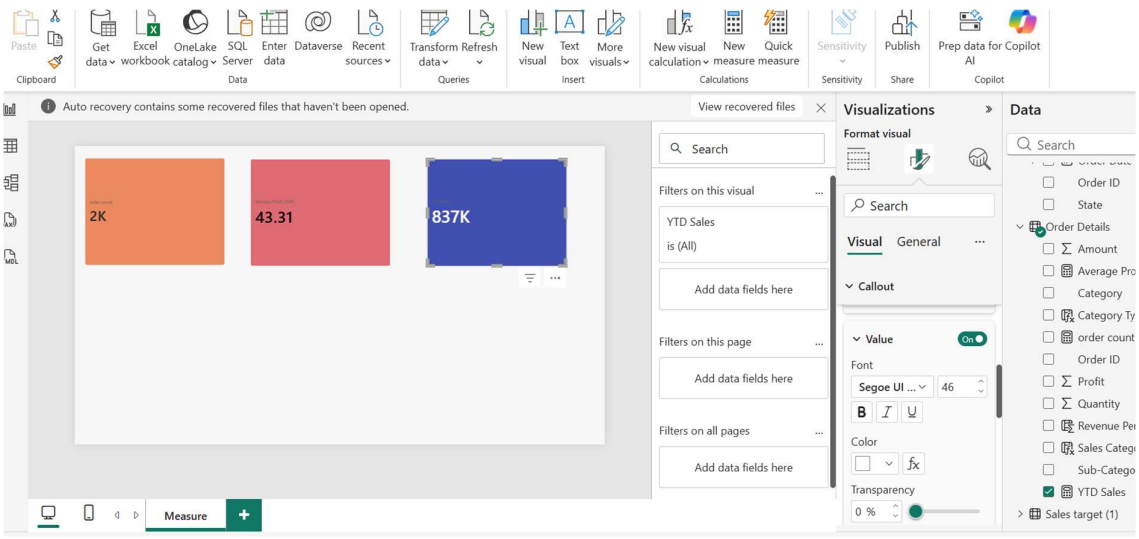
Calculate Average Profit in Delhi: Created a measure to calculate the average profit for orders placed in Delhi.





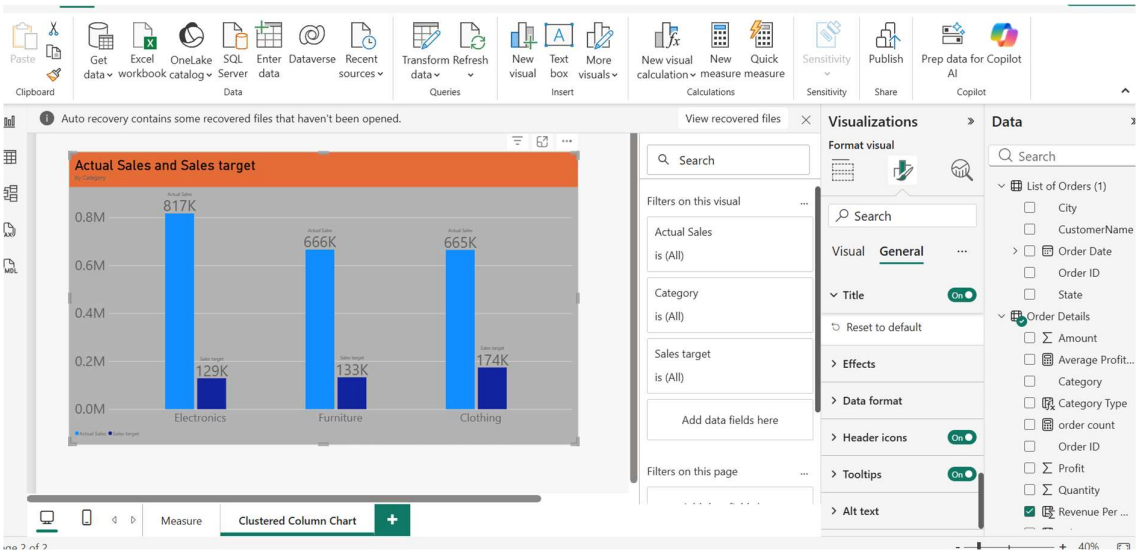
Calculate Year-to-Date (YTD) Sales: Defined a measure to calculate the total sales amount accumulated from the earliest order date up to each order date.



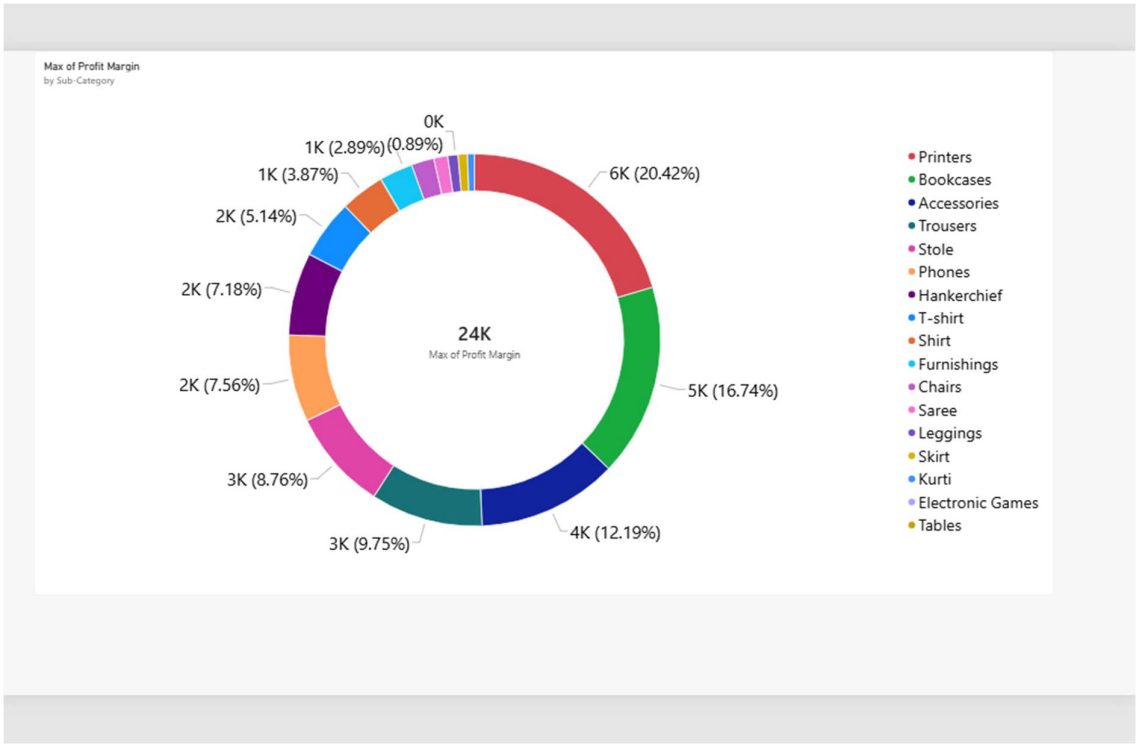


Data Visualization:

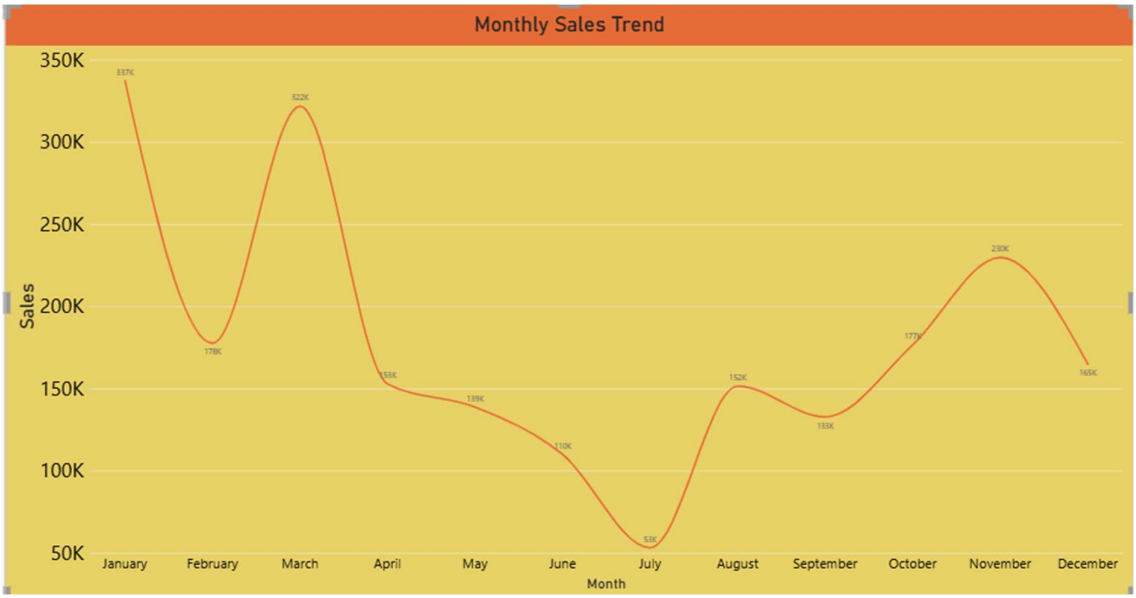
Sales Target Achievement by Category: Compared actual sales with sales targets by category using a clustered column chart



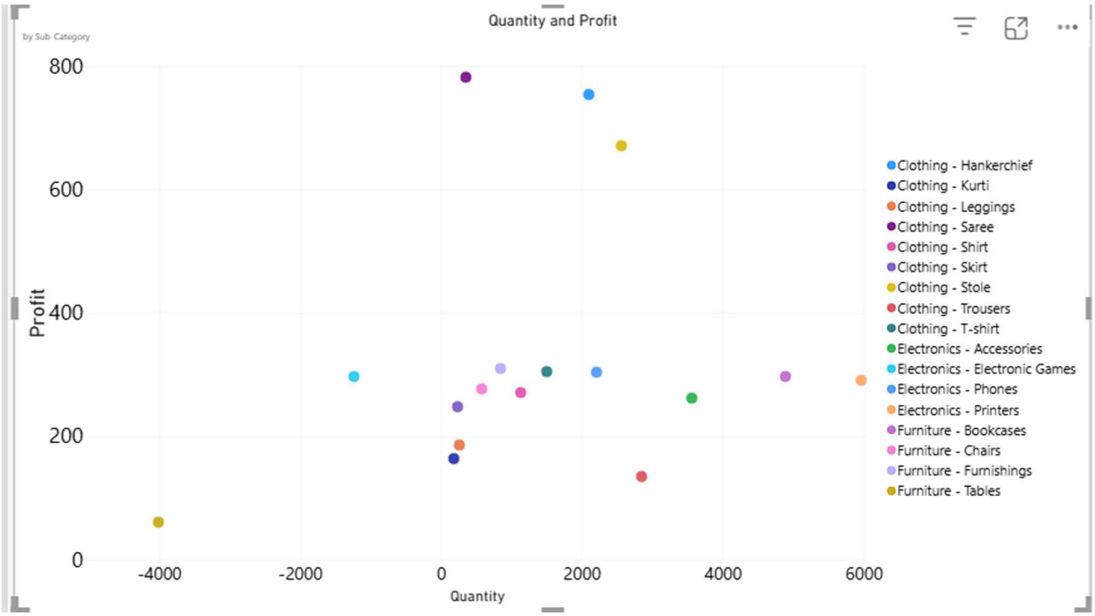
- Max Profit Margin by Sub-Category: Analyzed the maximum profit margin for each sub-category of products using a donut chart.



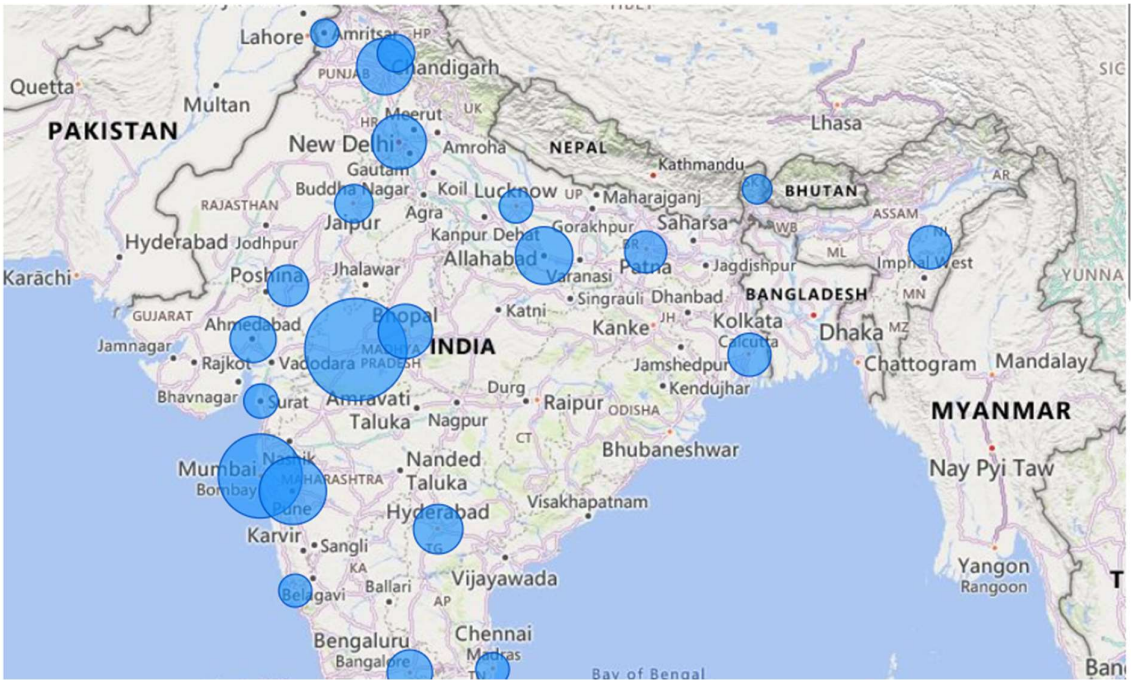
- Monthly Sales Trend: Displayed the trend of monthly sales over time using a line chart.



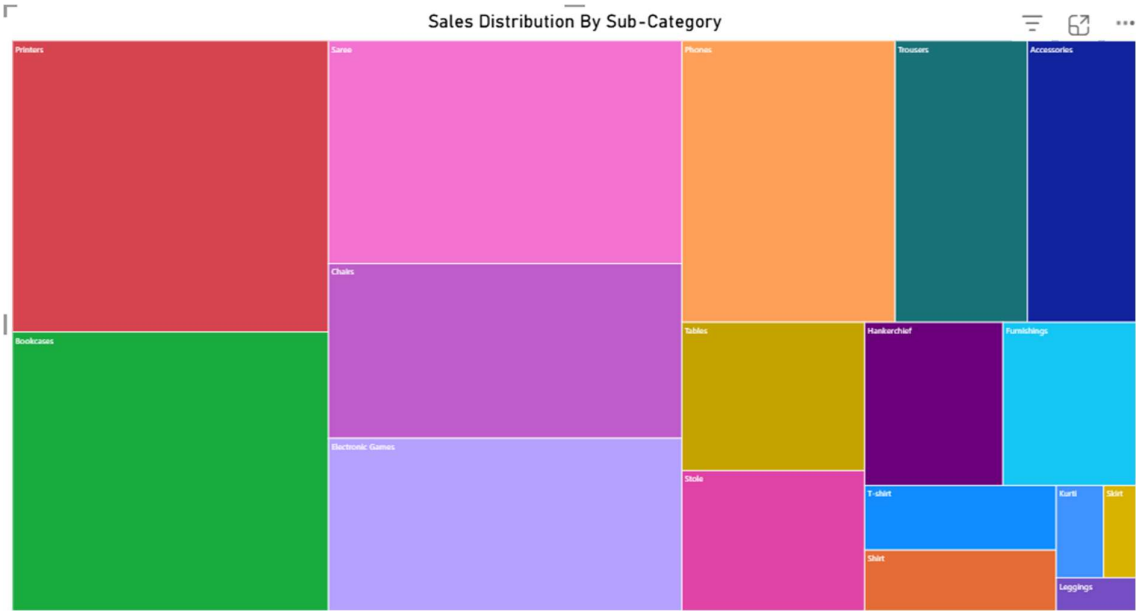
- Sub-Category: Compared the relationship between profit and quantity sold for different sub-categories using a scatter chart.



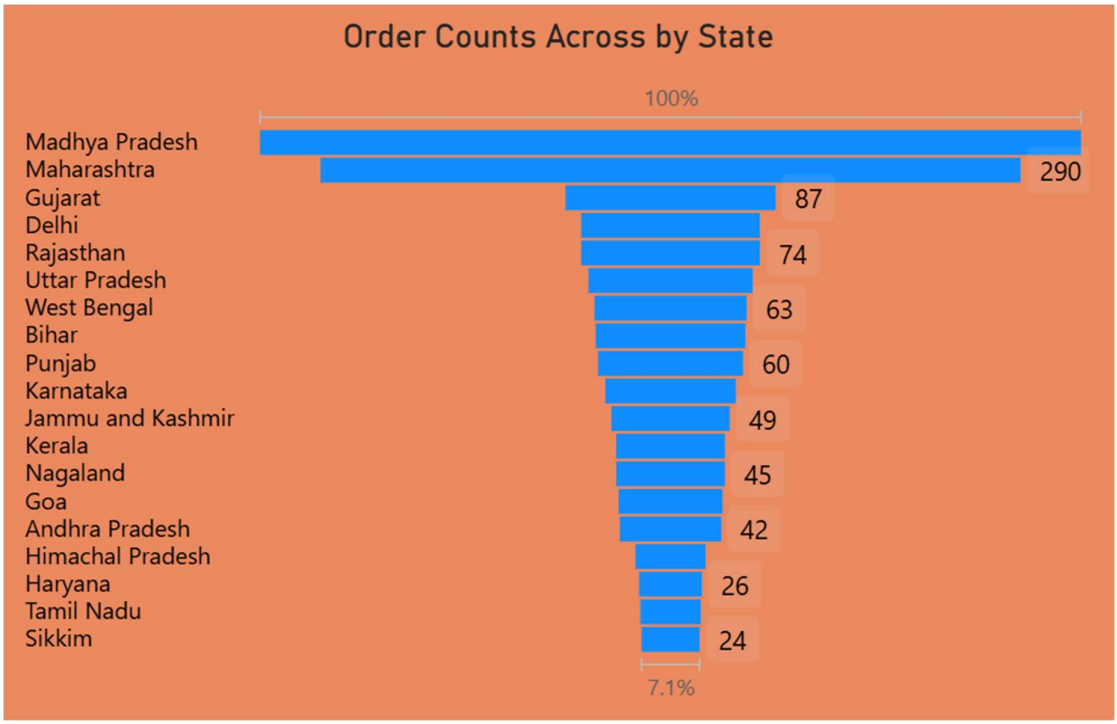
Geographic Sales Analysis: Displayed total sales on a map by city to identify regional sales patterns.



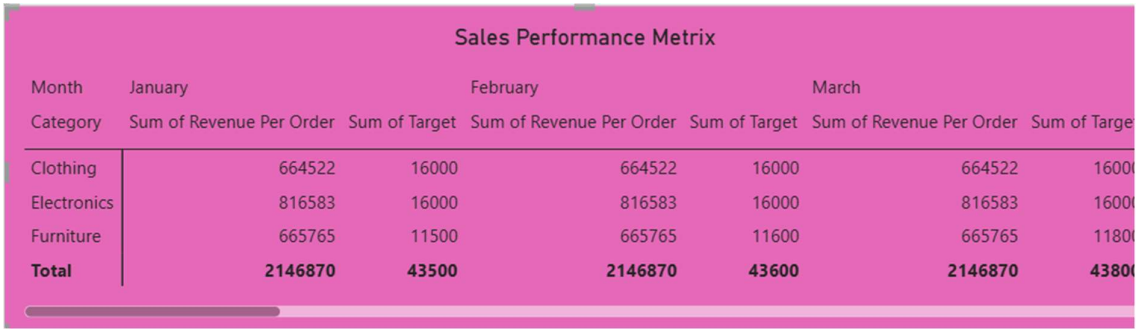
- Sales Distribution by Sub-Category: Represented the sales distribution across different sub-categories using a tree map.



- Order Count Analysis by State: Created a funnel chart to visualize the distribution of order counts across different states.



- Sales Performance Matrix: Created a matrix view to analyze how actual sales compare sales targets across different categories and months.



Sales Performance Metrix

Month	January		February		March	
Category	Sum of Revenue Per Order	Sum of Target	Sum of Revenue Per Order	Sum of Target	Sum of Revenue Per Order	Sum of Targe
Clothing	664522	16000	664522	16000	664522	1600
Electronics	816583	16000	816583	16000	816583	1600
Furniture	665765	11500	665765	11600	665765	1180
Total	**2146870**	**43500**	**2146870**	**43600**	**2146870**	**4380**

- Comparison of Total Sales Amount and Target: Created cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, created a multi-row card to display the minimum target for each segment.

